

# Optimal Trading Against a Signal: a Wiener-Hopf Approach

Matteo Smerlak\*

**Keywords.** Optimal execution; market impact; propagator model; Wiener-Hopf factorization; Gohberg-Krein factorization; fractional calculus; algorithmic trading.

**JEL classification.** C61; G11; G12; G14.

**AMS classification.** 45E10; 60G35; 91G80; 26A33.

## Abstract

We study the problem of maximizing expected gain from a return-predicting signal net of the impact cost of one's own trades, under a propagator impact model with a general symmetric positive-definite kernel. Existing treatments express the adapted optimal trading rate as the solution of an integral or stochastic Volterra equation. We give a closed-form bulk expression for this rate using a causal-anticausal (Wiener-Hopf) factorization of the impact cost operator. For the empirically supported power-law kernel  $G(t) = |t|^{-\beta}$  with  $\beta \in (0, 1)$ , the factors are fractional integrals, and the optimal trading policy simplifies to a composition of fractional derivatives of order  $(1 - \beta)/2$  applied to the signal and its forecasts. On a finite horizon the same recipe produces an optimal policy that converges to the whole-line formula in the interior at rate  $O(d(t)^{-(1-\beta)/2})$ , where  $d(t)$  is the distance from  $t$  to the nearest boundary.

## 1 Introduction

### 1.1 The gain-cost problem

Institutional trading of large orders — index rebalances, algorithmic strategies against short-horizon forecasts, hedges against derivative books — routinely moves quantities that are large multiples of top-of-book depth. Empirical studies of trade-and-quote data (Lillo et al. 2003, Bouchaud et al. 2004, Gatheral 2010, Jusselin and Rosenbaum 2020) establish that each child order pushes the mid-price against the trader's direction and that the resulting *transient impact* is well described by a translation-invariant propagator kernel  $G(t - s)$  that decays with lag: the price at time  $t$  carries a term  $\int_{s \leq t} G(t - s) u_s ds$  from the trader's own past rate  $u$ . The cost of a large parent order therefore depends on the whole path of its execution schedule, not only on its instantaneous rate.

When the trader holds a return-predicting signal  $\alpha_t$  — a forecast of price change conditional on time- $t$  information — the gain from trading in the signal's direction competes with the impact cost of moving into the position. Under the propagator model of Bouchaud, Gefen, Potters and Wyart (2004), extended by Gatheral (2010), the expected P&L of an adapted rate  $u \in L^2_{\text{adap}}$  is a linear gain against a quadratic cost, and the *gain-cost problem* is

$$\max_{u \in L^2_{\text{adap}}} \mathbb{E} \int u_t \alpha_t dt - \frac{\gamma}{2} \mathbb{E} \iint G(|t - v|) u_t u_v dt dv, \quad (1)$$

---

\*Capital Fund Management, 23 rue de l'Université, 75007 Paris, France. [matteo.smerlak@cfm.com](mailto:matteo.smerlak@cfm.com)

with  $\gamma > 0$  a cost-aversion coupling. Write  $C$  for the symmetric convolution  $(Cu)(t) = \int G(|t - v|)u_v dv$ , positive-definite ( $\hat{C}(\xi) \geq 0$ ) to rule out static round-trip arbitrage. Two kernels run through the paper: the empirically supported power law  $G(t) = |t|^{-\beta}$ ,  $\beta \in (0, 1)$  estimated at 0.2–0.6 across markets and asset classes (Lillo et al. 2003, Bouchaud et al. 2004, Gatheral 2010, Jusselin and Rosenbaum 2020), with symbol  $\hat{C}(\xi) = c_\beta |\xi|^{\beta-1}$  and  $c_\beta = 2\Gamma(1 - \beta) \sin(\pi\beta/2)$ ; and the tractable exponential  $G(t) = e^{-\kappa|t|}$ ,  $\kappa > 0$ , with symbol  $2\kappa/(\kappa^2 + \xi^2)$ . The power-law symbol has a low-frequency singularity and  $C$  is unbounded on  $L^2(\mathbb{R})$ ; §4 gives the technical domain fix required for the pure power-law setting. The exponential kernel and the finite-horizon truncation to  $[0, T]$  are bounded on  $L^2$  without further regularization.

## 1.2 Analogy with Markowitz portfolio choice

Problem (1) has the same convex-quadratic structure as the mean–variance portfolio problem (Markowitz 1952, Merton 1972),

$$\max_{w \in \mathbb{R}^N} w^\top \mu - \frac{1}{2} \lambda w^\top \Sigma w, \quad (2)$$

under the correspondence

$$(w, \mu, \Sigma, \lambda) \longleftrightarrow (u, \alpha, C, \gamma).$$

The Markowitz optimum is  $w^* = \lambda^{-1} \Sigma^{-1} \mu$  with value  $\frac{1}{2\lambda} \mu^\top \Sigma^{-1} \mu$  (the Mahalanobis norm of expected return, squared). The direct analog  $u^* = \gamma^{-1} C^{-1} \alpha$  would be the optimum of the trading problem before imposing adaptedness and accounting for the non-locality of the Hessian. Two features of the temporal problem block the direct analog:

(i) *Non-locality of the Hessian.*  $C$  is a temporal convolution against a symmetric kernel, so  $(Cu)(t)$  depends on  $u_v$  for  $v$  both before and after  $t$ .

(ii) *Adaptedness of the feasible set.* The trading rate  $u_t$  at time  $t$  must be  $\mathcal{F}_t$ -measurable; computing  $(C^{-1}\alpha)_t$  requires future values of  $\alpha$  which the trader does not have.

## 1.3 Prior treatments

Existing treatments of problem (1) establish existence, uniqueness, and structural characterizations of  $u^*$ .

*Deterministic constant-signal case.* Almgren and Chriss (2001) treat quadratic cost with the exponential kernel and derive closed-form schedules by state augmentation. Obizhaeva and Wang (2013) solve the same problem in a limit-order-book resilience framework. For the power-law kernel, Gatheral, Schied and Slynko (2012) give the optimal deterministic schedule as an eigenfunction of a Fredholm integral equation, exhibiting the Söhlgen–Tricomi endpoint weights  $(t(T - t))^{(\beta-1)/2}$ . In these papers the “signal” is a Lagrange multiplier fixed by the terminal-inventory constraint; the trader has no additional information.

*Stochastic-signal case, exponential kernel.* Lehalle and Neuman (2019) solve the Almgren–Chriss problem with a general adapted signal and linear temporary impact, obtaining  $u^*$  as a conditional expectation of the future signal — the temporary-impact cost operator is diagonal, so the projected inverse reduces to the projection itself. Neuman and Voß (2022) extend the analysis to the exponential resilience kernel, using the Markov property to reduce the problem to a finite-dimensional linear-quadratic control with Riccati feedback — available via state augmentation specific to the exponential kernel and not to the empirically supported power-law case. Cartea, Jaimungal and Penalva (2015) develop the broader algorithmic-trading framework.

*Stochastic-signal case, general kernel.* For a general propagator kernel with a stochastic signal, Abi Jaber and Neuman (2025) prove existence and uniqueness of  $u^*$  and characterize it as the unique solution of a linear stochastic Volterra equation of the second kind: a fixed-point equation involving the kernel and the signal, from which closed forms are recovered only after kernel-specific reductions. The framework extends to matrix-valued cross-impact (Abi Jaber et al. 2024b) and constrained trading including battery-storage applications (Abi Jaber et al. 2024a), inheriting the same Volterra-equation form.

*Stochastic-signal case, power-law kernel.* Forde, Sánchez-Betancourt and Smith (2022) treat Gaussian signals on  $[0, T]$  using the finite-interval factorization  $G_T = T T^*$  with  $T$  a Volterra operator of Riemann–Liouville type (Porter and Stirling 1990, Ex. 6.2, 9.2); the optimal rate is the integral of the signal against the resolvent of  $T$ , inverted via the Chakrabarti–George Abel formula (Chakrabarti and George 1994). The representation is horizon-tied and gives no bulk formula in terms of the signal path.

*Deterministic factorization theory.* Wiener and Hopf (1931) introduced the factorization of scalar Fourier symbols on a half-line as the basis for solving convolution integral equations. Krein (1962) developed the corresponding half-line integral-equation theory, and Wiener (1949) applied the factorization to stationary linear prediction. Gohberg and Krein (1970) established the finite-interval factorization  $I + K = (I + L^*)(I + L)$  for positive kernels via Volterra triangularization, and Arveson (1975) generalized the picture to positive operators on Hilbert spaces equipped with a continuous nest of projections.

Across these strands the adapted optimal rate under a stochastic signal and a power-law kernel is characterized implicitly, through fixed-point Volterra equations or horizon-tied resolvent representations. No bulk closed form in terms of the signal path has been given, and the fractional-calculus structure of the problem is left buried in the resolvent. The factorization identity below closes both gaps.

## 1.4 Contribution

We give a closed formula for the adapted optimal trading rate under a general symmetric positive-definite propagator kernel, and show that for the power-law kernel it reduces in the interior of the trading horizon to a fractional derivative of the trader’s forecast curve.

**Closed formula.** For any factorization  $C = C_- C_+$  of the cost operator into a causal Volterra factor  $C_+$  and its adjoint  $C_-$ , the adapted first-order condition of (1) admits the solution

$$u^* = \gamma^{-1} C_+^{-1} P_+ C_-^{-1} \alpha, \quad (3)$$

via the projected-inverse identity from nest-algebra theory (Arveson 1975),

$$(P_+ C P_+)^{-1} = C_+^{-1} P_+ C_-^{-1} \quad (4)$$

on the adapted subspace  $L_{\text{adap}}^2$ . The factorization is Wiener–Hopf on the whole line and Gohberg–Krein on a finite interval, and (3) closes the problem in both cases.

The three composed operators act on the signal: the anticausal square-root inverse  $C_-^{-1}$  applied to  $\alpha$  produces an  $\mathcal{F}_s$ -measurable “whitened” process at each  $s$  once future signal values are replaced by their conditional expectations (the *forecast curve*  $\bar{\alpha}(t, s) = \mathbb{E}_t[\alpha_s]$ ); the projection  $P_+$  enforces adaptedness; the causal square-root inverse  $C_+^{-1}$  colors the whitened process back to obtain the trading rate. The three-step architecture — anticausal whitening, adapted projection, causal coloring — is the temporal analog of the Wiener–Kolmogorov prediction filter for stationary processes (Wiener 1949), with  $C_+$  in place of the process spectral square root.

**Power-law kernel: bulk formula.** For  $G(t) = |t|^{-\beta}$  the whole-line factors are the Liouville (Weyl) fractional integrals  $C_{\pm} = c_{\beta}^{1/2} I_{\pm}^{\nu}$  with  $\nu = (1 - \beta)/2$ , and (3) reduces to a fractional derivative applied to the forecast curve:

$$u_t^* = \gamma^{-1} c_{\beta}^{-1} (D_+^{\nu} \zeta)(t), \quad \zeta_s = (D_-^{\nu} \bar{\alpha}(s, \cdot))(s). \quad (5)$$

The optimal rate at time  $t$  is the causal Marchaud fractional derivative, of order  $\nu = (1 - \beta)/2$ , of the “whitened signal”  $\zeta_s$ ; and  $\zeta_s$  is the anticausal Marchaud fractional derivative of the same order applied to the forecast curve  $\bar{\alpha}(s, \cdot)$  and evaluated at the diagonal. Without the adapted projection the composition  $D_+^{\nu} D_-^{\nu}$  would collapse (in the frequency domain) to the symmetric Riesz derivative of order  $2\nu = 1 - \beta$ ; the projection  $P_+$  separates the two half-order pieces and prevents that collapse, so the operator  $D_+^{\nu} P_+ D_-^{\nu}$  is genuinely a composition of one-sided derivatives of order  $\nu$  rather than a single fractional derivative. The empirical range  $\beta \in (0.2, 0.6)$  places each half-order in  $(0.2, 0.4)$ .

## 2 The Gain–Cost Problem in Operator Form

### 2.1 Setup

Fix a filtered probability space  $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}}, \mathbb{P})$  satisfying the usual conditions. The signal  $\alpha$  is a mean-zero, progressively measurable process with stationary spectral density  $S_{\alpha}(\xi)$  satisfying  $\int (1 + |\xi|^{2(1-\beta)+\epsilon}) S_{\alpha}(\xi) d\xi < \infty$  for some  $\epsilon > 0$ . Let  $L^2 = L^2(\Omega \times \mathbb{R})$  with reference measure  $\mathbb{P} \otimes dt$ , and let  $L_{\text{adap}}^2 \subset L^2$  denote the closed subspace of adapted processes: those  $u$  with  $u_t \in \mathcal{F}_t$  for a.e.  $t$ . Denote by  $P_+$  the  $L^2$ -orthogonal projection onto  $L_{\text{adap}}^2$ ; pointwise,

$$(P_+ X)_s = \mathbb{E}_s[X_s],$$

which coincides with the optional projection restricted to  $L^2$ . Our Fourier convention is  $\hat{f}(\xi) = \int e^{i\xi t} f(t) dt$ ; a causal function (support in  $t \geq 0$ ) has Fourier transform analytic in the upper half-plane.

The cost operator  $C$  is the deterministic convolution against a symmetric kernel  $G$ ,

$$(Cu)(t) = \int G(|t - v|) u_v dv, \quad \hat{C}(\xi) \geq 0 \text{ on } \xi \neq 0,$$

acting on  $L^2$  diagonally in  $\omega$ . Positive-definiteness of  $\hat{C}$  ensures strict convexity of the quadratic penalty in (1) on  $L_{\text{adap}}^2$ . For the pure power-law kernel  $C$  is unbounded on  $L^2(\mathbb{R})$  and is interpreted between homogeneous Sobolev spaces  $C: \dot{H}^{-\nu} \rightarrow \dot{H}^{\nu}$  with  $\nu = (1 - \beta)/2$ ; the technical domain fix is developed in §4. On  $[0, T]$  the kernel is weakly singular and  $G_T$  is bounded on  $L^2([0, T])$  without further regularization.

The *forecast curve* at time  $t$  is

$$\bar{\alpha}(t, s) = \begin{cases} \alpha_s, & s \leq t, \\ \mathbb{E}_t[\alpha_s], & s > t. \end{cases} \quad (6)$$

### 2.2 The adapted first-order condition

Taking the Gâteaux derivative of the objective in (1) against adapted variations  $\delta u \in L_{\text{adap}}^2$  and using  $\mathbb{E}[u \mathbb{E}_t[\cdot]] = \mathbb{E}[u \cdot]$  for adapted  $u$  gives the *adapted first-order condition*

$$\gamma \mathbb{E}_t[(Cu^*)(t)] = \alpha_t \text{ for a.e. } t. \quad (7)$$

Equivalently in operator form:  $\gamma P_+ C P_+ u^* = \alpha$  with  $u^*, \alpha \in L_{\text{adap}}^2$ .

The Hessian  $\gamma P_+ C P_+$  is a symmetric strictly positive operator on  $L_{\text{adap}}^2$  (strict convexity follows from  $\hat{C} > 0$  on  $\xi \neq 0$  via Plancherel applied to the extension by zero of any adapted  $u$ ). It admits a bounded two-sided inverse, and (7) has a unique solution

$$u^* = \gamma^{-1} (P_+ C P_+)^{-1} \alpha. \quad (8)$$

The remainder of the paper computes the projected inverse in closed form.

### 3 Factorization of the Cost Operator

The strategy is to split  $C$  as a causal-anticausal product and reduce the projected inverse to a three-step alternating operator. This section states the factorization in both settings — whole line and finite interval — and proves the projected-inverse identity that underlies the closed-form solution.

#### 3.1 Whole line: Wiener–Hopf factorization

**Proposition 1** (Wiener–Hopf factorization). *Let  $\hat{C}(\xi) \geq 0$  with  $\log \hat{C}(\xi)/(1 + \xi^2) \in L^1(\mathbb{R})$ . Then  $\hat{C}$  admits a factorization*

$$\hat{C}(\xi) = \hat{C}_-(\xi) \hat{C}_+(\xi), \quad (9)$$

*with  $\hat{C}_+$  analytic and non-vanishing in the closed upper half-plane,  $\hat{C}_-$  analytic in the closed lower half-plane, and  $\hat{C}_+(\xi) = \hat{C}_-(\xi)$ ; equivalently  $C_+^* = C_-$ . In the time domain,  $C_+$  has kernel supported on  $\{s \leq t\}$  (causal Volterra) and  $C_-$  on  $\{s \geq t\}$  (anticausal). Under the normalization  $\hat{C}_+ = \hat{C}_-$  the factorization is unique up to a unimodular multiplicative constant.*

Existence is classical (Wiener and Hopf 1931, Krein 1962). Kernel-specific factors are worked out in §4 (power-law) and §5.1 (exponential).

#### 3.2 Finite interval: Gohberg–Krein factorization

On  $[0, T]$  translation invariance is lost, Fourier factorization does not apply, and the appropriate analog is the Volterra-triangular factorization of Gohberg and Krein (1970), abstractly the Arveson outer factorization on a continuous nest (Arveson 1975).

**Proposition 2** (Gohberg–Krein factorization). *Let  $G_T$  be the restriction of  $C$  to  $L^2([0, T])$ , viewed as an integral operator with weakly singular kernel  $G(|t - v|)\mathbf{1}_{[0, T]^2}$ , and suppose  $G_T$  is strictly positive on  $L^2([0, T])$  (injective with  $\langle G_T f, f \rangle > 0$  for  $f \neq 0$ , guaranteed by  $\hat{C} > 0$  on  $\xi \neq 0$  via Plancherel on the extension by zero). Then  $G_T$  admits a causal-anticausal factorization*

$$G_T = C_- C_+, \quad (10)$$

*with  $C_+$  a Volterra operator whose kernel is supported on  $\{s \leq t\} \cap [0, T]^2$  (causal) and  $C_- = C_+^*$  its adjoint (kernel on  $\{s \geq t\}$ ). The causal factor  $C_+$  is unique up to a left unitary commuting with the nest  $\{P_{[0, t]}\}$  (a diagonal phase). For the power-law kernel  $G_T$  is compact with unbounded inverse, so the factors act between weighted Sobolev spaces rather than on  $L^2$  (§4.1).*

The classical Gohberg–Krein outer factor  $T$  realizes the Arveson outer factorization of  $G_T$  with respect to the continuous nest  $\{P_{[0,t]}\}_{t \in [0,T]}$  as  $G_T = TT^*$  with  $T$  causal (Arveson 1975, Thm 4.4.2)(Porter and Stirling 1990, Ex. 6.2/9.2). This places the causal factor on the *left*; the adapted projected inverse of Lemma 1 (below) instead needs the causal factor on the right, obtained by the time reflection  $C_+ = RT^*R$ ,  $(Rf)(t) = f(T - t)$ , since  $RG_T R = G_T$  gives  $C_-C_+ = R(TT^*)R = G_T$ . The explicit power-law factors are given in §4.

### 3.3 The adapted projected inverse

**Lemma 1** (Adapted inverse via causal-anticausal factorization). *Let  $\mathcal{H}$  be either  $L^2(\mathbb{R})$  or  $L^2([0, T])$ , tensored with  $L^2(\Omega)$  against the filtered measure. Let  $C = C_-C_+$  be a factorization of a symmetric strictly positive operator on  $\mathcal{H}$  into a causal Volterra factor  $C_+$  (kernel supported on  $\{s \leq t\}$ ) and its adjoint  $C_- = C_+^*$ . Let  $P_+$  be the  $L^2$ -orthogonal projection onto adapted processes. Then on the adapted subspace,*

$$(P_+ C P_+)^{-1} = C_+^{-1} P_+ C_-^{-1}. \quad (11)$$

*Proof (sketch).* Causality of  $C_+$  (kernel on  $\{s \leq t\}$ ) gives  $P_+^\perp C_+ P_+ = 0$ , so  $C_+^{-1}$  preserves the adapted subspace; its adjoint identity is  $P_+ C_- P_+^\perp = 0$ , so  $C_-$  preserves the anticausal subspace. Splitting  $C_-^{-1}u$  into adapted and anticausal parts and using the second identity to kill the anticausal contribution, one checks that for adapted  $u$ ,  $P_+ C_-^{-1} P_+ C_- u = P_+ u = u$ . Composing with  $C_+^{-1}$  on the left and  $P_+ C P_+$  on the right returns  $u$ . Only causality of  $C_+$  and self-adjointness of  $C$  enter, so the identity covers the whole-line Wiener–Hopf and the finite-interval Gohberg–Krein/Arveson factorizations. Full details in the appendix.  $\square$

### 3.4 Closed-form optimal trading rate

Combining (8) and (11):

**Theorem 1** (Closed-form optimal rate). *Under the hypotheses of §2.1, the unique adapted solution of (7) is*

$$u^* = \gamma^{-1} C_+^{-1} P_+ C_-^{-1} \alpha, \quad (12)$$

*understood in either setting: on the whole line with  $C_\pm$  the Wiener–Hopf factors, or on  $[0, T]$  with  $C_+ = RT^*R$  the reflected (terminal-anchored) causal Gohberg–Krein factor of  $G_T$  and  $C_- = C_+^*$ . When linear position constraints are imposed,  $\alpha$  is replaced by an effective signal  $\alpha^{\text{eff}} = \alpha + \sum_k \mu_k \psi_k$  adjoining one multiplier per constraint: a finite family for expectation constraints such as  $\mathbb{E}[X_T] = 0$ , and a process-valued multiplier  $\mu(\cdot)$  for the pathwise liquidation constraint  $X_T = 0$  a.s. (Forde et al. 2022).*

The acausal factor  $C_-^{-1}$  applied to the adapted signal  $\alpha$  at time  $s$  requires the future path  $\{\alpha_r\}_{r \geq s}$ , which is not in  $\mathcal{F}_s$ . The adapted projection replaces this future by its conditional expectation, and commutation of the deterministic operator  $C_-^{-1}$  with conditional expectation on the argument variable (Hytönen et al. 2016, Prop. 2.6.13) gives

$$(P_+ C_-^{-1} \alpha)_s = (C_-^{-1} \bar{\alpha}(s, \cdot))(s). \quad (13)$$

The optimal rate is therefore

$$u_t^* = \gamma^{-1} C_+^{-1} \zeta|_t, \quad \zeta_s = (C_-^{-1} \bar{\alpha}(s, \cdot))(s). \quad (14)$$

The forecast curve enters only through the inner  $C_-^{-1}$  step;  $C_+^{-1}$  then samples the adapted process  $\zeta$  over  $s \leq t$ , producing an  $\mathcal{F}_t$ -measurable rate.

## 4 Power-Law Kernel

For  $G(t) = |t|^{-\beta}$  with  $\beta \in (0, 1)$  and  $\nu = (1 - \beta)/2$ , the Wiener–Hopf and Gohberg–Krein factors are explicit fractional integrals, and Theorem 1 reduces to a fractional derivative of the forecast curve on the whole line and to a boundary-deformed fractional derivative on  $[0, T]$ .

### 4.1 Fractional-integral factors

*Whole-line factors.* The symbol  $\hat{C}(\xi) = c_\beta |\xi|^{\beta-1} = c_\beta (i\xi)^{-\nu} (-i\xi)^{-\nu}$  (standard branches) factors as  $\hat{C}_+(\xi) = c_\beta^{1/2} (-i\xi)^{-\nu}$  (analytic in the upper half-plane) and  $\hat{C}_-(\xi) = c_\beta^{1/2} (i\xi)^{-\nu}$ . In the time domain,  $C_\pm$  are the causal and anticausal Riemann–Liouville fractional integrals of order  $\nu$ :

$$C_+ = c_\beta^{1/2} I_+^\nu, \quad (I_+^\nu f)(t) = \frac{1}{\Gamma(\nu)} \int_{-\infty}^t (t-s)^{\nu-1} f(s) ds; \quad C_- = c_\beta^{1/2} I_-^\nu. \quad (15)$$

Inverses are the Marchaud fractional derivatives  $C_\pm^{-1} = c_\beta^{-1/2} D_\pm^\nu$  (Samko et al. 1993, §5.4).

*Boundedness caveat.* The pure power-law symbol satisfies the log-integrability hypothesis of Proposition 1, so the symbol factorization (9) exists distributionally. The factors are however unbounded on  $L^2(\mathbb{R})$ :  $\hat{C}_\pm(\xi)$  diverges at  $\xi = 0$  and  $\hat{C}_\pm^{-1}$  diverges as  $|\xi| \rightarrow \infty$ , so the projected-inverse identity of §3.3 is an equality between unbounded operators. It holds on the dense domain fixed by the standing spectral hypothesis of §2.1, which ensures  $C^{-1}\alpha \in L^2$  and gives sense to each Marchaud composition. A bounded-operator setup is recovered by adjoining a temporary-impact term  $\frac{1}{2}\eta u_t^2$  (§5.2): the modified symbol  $M(\xi) = c_\beta |\xi|^{\beta-1} + \eta/\gamma$  has  $M^{-1} \in L^\infty(\mathbb{R})$ , Proposition 1 applies with  $C_\pm^{-1}$  bounded on  $L^2$  without spectral-decay assumptions on  $\alpha$ , and the  $\eta \rightarrow 0$  limit recovers the pure power-law formulas on the §2.1 domain.

*Finite-interval factors.* On  $[0, T]$  the classical Gohberg–Krein outer construction (Forde et al. 2022, Porter and Stirling 1990) gives  $G_T = T T^*$  with

$$(T\varphi)(t) = \int_0^t k(s, t) \varphi(s) ds, \quad k(s, t) = c_\beta^{1/2} (t/s)^\nu (t-s)^{\nu-1} / \Gamma(\nu), \quad (16)$$

equivalently  $T = c_\beta^{1/2} B^{-1} I_+^\nu B$  with  $B(t) = t^{-\nu}$ , the left-endpoint-anchored form. This outer factorization places the causal factor on the *left*; the adapted projected inverse of Lemma 1 requires the causal factor on the right,  $G_T = C_- C_+$ . By the reflection  $(Rf)(t) = f(T-t)$  and  $R G_T R = G_T$ , that factor is the terminal-anchored causal Volterra operator  $C_+ = R T^* R$ ,

$$C_+ = c_\beta^{1/2} B_T I_+^\nu B_T^{-1}, \quad C_- = C_+^* = c_\beta^{1/2} B_T^{-1} I_-^\nu B_T, \quad B_T(t) = (T-t)^{-\nu}, \quad (17)$$

with kernel  $c_\beta^{1/2} ((T-s)/(T-t))^\nu (t-s)^{\nu-1} / \Gamma(\nu)$  on  $\{s \leq t\}$ , so that  $G_T = C_- C_+ = c_\beta B_T^{-1} I_-^\nu B_T^2 I_+^\nu B_T^{-1}$  (verified by direct kernel integration). Both factors act boundedly between the weighted Sobolev spaces of Porter–Stirling (Porter and Stirling 1990, Ex. 6.2/9.2) and Samko–Kilbas–Marichev (Samko et al. 1993, §13.5). The terminal-anchored weight  $B_T$  absorbs the loss of translation invariance at the forward endpoint and pins  $C_+$  to the nest  $\{P_{[0,t]}\}$  up to a diagonal phase, breaking manifest  $t \leftrightarrow T-t$  symmetry while  $G_T = C_- C_+$  preserves it.

### 4.2 Bulk formula on the whole line

Substituting  $C_\pm^{-1} = c_\beta^{-1/2} D_\pm^\nu$  into (14):

**Corollary 1** (Power-law bulk formula). *For  $G(t) = |t|^{-\beta}$  with  $\beta \in (0, 1)$ ,  $\nu = (1 - \beta)/2$ , and  $\alpha$  stationary adapted satisfying the standing spectral hypothesis of §2.1,*

$$u_t^* = \gamma^{-1} c_\beta^{-1} (D_+^\nu \zeta)(t), \quad \zeta_s = (D_-^\nu \bar{\alpha}(s, \cdot))(s). \quad (18)$$

The inner fractional differentiation  $D_-^\nu$  acts on the forecast curve and cancels the frequency dependence of the impact operator; the outer  $D_+^\nu$  colors the resulting adapted process causally. Only in the anticipative case (future signal known, no projection) would the whitened process  $C_-^{-1} \alpha$  carry the flat-in-impact spectrum  $c_\beta^{-1} |\xi|^{1-\beta} S_\alpha(\xi)$ ; the adapted  $\zeta_s = (D_-^\nu \bar{\alpha}(s, \cdot))(s)$  replaces future values by forecasts and has a strictly smaller spectrum (for OU,  $\zeta_s = \theta^\nu \alpha_s$  with spectrum  $\theta^{2\nu} S_\alpha(\xi)$ , cf. (19)). Without the projection the composition  $D_+^\nu D_-^\nu$  would reduce (in symbol form) to the symmetric Riesz derivative  $|\xi|^{1-\beta}$ ; adaptedness splits it into the two one-sided derivatives of order  $\nu$  separated by  $P_+$ .

For OU  $\alpha$  with mean-reversion rate  $\theta$ , direct Marchaud integration against the exponential forecast tail gives  $(D_-^\nu \bar{\alpha}(t, \cdot))(t) = \theta^\nu \alpha_t$ , so

$$u_t^{*,\text{OU}} = \gamma^{-1} c_\beta^{-1} \theta^\nu (D_+^\nu \alpha)(t), \quad \mathbb{E}[u_t^{*,\text{OU}} \mid \alpha_t] = \gamma^{-1} c_\beta^{-1} \theta^{1-\beta} \alpha_t, \quad (19)$$

positive for every  $\beta \in (0, 1)$  and every  $\theta > 0$ : the scale-free causal inverse has no zero on the imaginary axis, so no sign-flip phase transition occurs as  $\theta$  varies (contrast the exponential kernel, §5.1). The OU signal satisfies the standing spectral hypothesis of §2.1 for  $\beta > 1/2$ ; for  $\beta \leq 1/2$  the OU spectrum  $\sigma^2/(\theta^2 + \xi^2)$  fails high-frequency integrability against  $|\xi|^{2(1-\beta)+\epsilon}$ , and (19) is made rigorous by adding a temporary-impact regularization (§5.2) or by working with increments of  $\alpha$ .

### 4.3 Boundary-deformed formula on a finite interval

Substituting (12) into Theorem 1:

**Corollary 2** (Power-law finite-interval formula). *For  $G(t) = |t|^{-\beta}$  on  $[0, T]$ , the adapted optimal rate is*

$$u_t^{*,T} = \gamma^{-1} c_\beta^{-1} (B_T D_+^\nu B_T^{-1} P_+ B_T^{-1} D_-^\nu B_T \alpha^{\text{eff}})(t), \quad (20)$$

where  $B_T(t) = (T - t)^{-\nu}$  is the terminal-anchored endpoint weight from the factors (17) (the multiplications by  $B_T$  commute with  $P_+$  since  $B_T$  is deterministic, so  $B_T^{-1} P_+ B_T^{-1} = P_+ B_T^{-2}$ ) and  $\alpha^{\text{eff}} = \alpha + \sum_k \mu_k \psi_k$  is the effective signal adjoining a multiplier per position constraint — a finite linear system for expectation constraints, and a process-valued multiplier for the pathwise liquidation constraint  $X_T = 0$  a.s. (Forde et al. 2022).

The composition applies, in order: (i) terminal weight  $B_T$  to the effective signal; (ii) anticausal Marchaud derivative  $D_-^\nu$ ; (iii) reweighting by  $B_T^{-1}$ ; (iv) adapted projection; (v) reweighting by  $B_T^{-1}$ ; (vi) causal Marchaud derivative  $D_+^\nu$ ; (vii) terminal weight  $B_T$ . The multiplications arise from substituting the reflected Gohberg–Krein factors  $C_\pm$  of (17) into  $u^* = \gamma^{-1} C_+^{-1} P_+ C_-^{-1} \alpha^{\text{eff}}$ . Forde, Sánchez-Betancourt and Smith (Forde et al. 2022, Thm 2.2) solve the same finite-interval problem through the full inverse  $G_T^{-1} = (T^*)^{-1} T^{-1}$ , for which the factor order is immaterial; (20) instead exposes the adapted projection between the two half-order factors, which requires the causal factor on the right.

Formula (20) is a *boundary-deformed fractional derivative*: the whole-line composition  $D_+^\nu P_+ D_-^\nu$  of (18) is preserved as the operator core; the terminal weights  $B_T, B_T^{-1}$  sandwich it, while the left endpoint enters through the truncation of the Marchaud kernels to  $[0, T]$ .



## 4.4 Interior asymptotic and rate of convergence

As  $T \rightarrow \infty$ , (20) converges pointwise to (18) in the interior. Let  $d(t) = \min(t, T - t)$  be the distance from  $t$  to the nearest boundary. Centering at the midpoint  $t = T/2 + t'$ , the terminal weight  $B_T(T/2 + t') = (T/2 - t')^{-\nu}$  factors as a  $T$ -dependent scalar  $(T/2)^{-\nu}$  times a slowly varying interior factor  $(1 - 2t'/T)^{-\nu}$ . The scalar cancels in the  $B_T^{-1} \cdot B_T$  conjugations of (20), and the interior factor converges to 1 uniformly on compact  $t'$  sets, with relative deviation  $O(d(t)^{-1})$ . The reflected Volterra kernel  $((T-s)/(T-t))^\nu (t-s)^{\nu-1}$  satisfies  $((T-s)/(T-t))^\nu \rightarrow 1$  in the interior, converging to the whole-line causal Riemann–Liouville kernel  $(t-s)^{\nu-1}/\Gamma(\nu)$ .

**Proposition 3** (Interior error bound). *For  $\alpha$  bounded and effective signal  $\alpha^{\text{eff}} \in \dot{H}^\nu(\mathbb{R})$ ,*

$$|u_t^{\star, T} - u_t^{\star, \mathbb{R}}| \leq C_1(\beta) \|\alpha\|_{L^\infty} d(t)^{-\nu} + C_2(\beta) \|\alpha^{\text{eff}}\|_{\dot{H}^\nu} T^{-\nu} d(t)^{-\nu}, \quad (21)$$

with  $\nu = (1 - \beta)/2$  and  $C_1(\beta), C_2(\beta)$   $\beta$ -dependent constants.

The first term is the Marchaud truncation error from cutting the fractional-derivative tail at the boundary; it dominates the weight-conjugation deviation, which is  $O(d(t)^{-1})$  and hence subdominant for  $\nu < 1$ . The second term is the endpoint-constraint correction, entering through the Söngen–Tricomi eigenfunction  $\phi_1(t) = [t(T-t)]^{(\beta-1)/2}$  of the no-signal solution (Gatheral et al. 2012, Forde et al. 2022), whose coefficient is  $O(\|\alpha^{\text{eff}}\|_{\dot{H}^\nu} T^{-\nu})$  under the stated regularity. For empirical  $\beta \in (0.2, 0.6)$ ,  $\nu \in (0.2, 0.4)$ , giving slow  $d(t)^{-\nu}$  interior convergence — a quantitative expression of the long memory of the impact kernel.

## 5 Discussion

### 5.1 Exponential kernel

The exponential kernel  $G(t) = e^{-\kappa|t|}$  is a tractable alternative to the power-law. It bounds the cost operator on  $L^2(\mathbb{R})$  and underlies the Obizhaeva–Wang (2013) resilience model and much of the Almgren–Chriss line of work. Working through it in the present framework does two things: it removes the domain subtleties that arise for the power-law kernel in §4, and it makes visible which features of Theorem 1 are structural (the three-step factorization) and which are specific to the fractional case (unbounded factors, fractional-derivative discretization, absence of a sign-flip regime).

The symbol

$$\hat{C}(\xi) = \frac{2\kappa}{\kappa^2 + \xi^2} = \frac{\sqrt{2\kappa}}{\kappa - i\xi} \cdot \frac{\sqrt{2\kappa}}{\kappa + i\xi}$$

splits into rational one-sided factors that are bounded on  $L^2(\mathbb{R})$ . Their inverses are first-order differential operators,

$$C_+^{-1} = (2\kappa)^{-1/2}(\kappa + \partial_t), \quad C_-^{-1} = (2\kappa)^{-1/2}(\kappa - \partial_t),$$

and substituting into Theorem 1 gives the bulk formula

$$u_t^{\star, \text{exp}} = \frac{1}{2\kappa\gamma} (\kappa + \partial_t) \zeta_t, \quad \zeta_s = (\kappa - \partial_r) \bar{\alpha}(s, r) \Big|_{r=s+}. \quad (22)$$

The optimal rate is a local first-order combination of the forecast curve and its time derivative — a striking simplification of the general nonlocal recipe, made possible by the rational symbol.

For an Ornstein–Uhlenbeck signal with mean reversion  $\theta$ , the whitened process reduces to  $\zeta_s = (\kappa + \theta)\alpha_s$ , and (22) yields

$$u_t^{\star, \text{exp}} = \frac{\kappa + \theta}{2\kappa\gamma} [(\kappa - \theta)\alpha_t + \sigma\dot{W}_t], \quad \mathbb{E}[u_t^{\star, \text{exp}} \mid \alpha_t] = \frac{\kappa^2 - \theta^2}{2\kappa\gamma} \alpha_t. \quad (23)$$

The white-noise term  $\sigma\dot{W}_t$  reflects that the exponential-kernel cost does not penalize the high-frequency content of the rate, so  $u^{\star, \text{exp}}$  is a distribution-valued optimum; it is regularized to an  $L^2$  process by any temporary-impact term  $\frac{1}{2}\eta u_t^2$  (§5.2), the empirically relevant case. The conditional mean rate flips sign at  $\theta = \kappa$ . When the signal mean-reverts faster than the impact tail decays ( $\theta > \kappa$ ), the trader takes the opposite side of the signal on average: any signal-following trade would leave an impact residual longer-lived than the signal itself, so acting against the signal amortises the cost. The power-law analog (19) shows no such transition; the scale-free causal inverse has a branch point at  $\xi = 0$  and no zero on the imaginary axis, so no pole–zero crossing occurs as  $\theta$  varies.

## 5.2 Temporary impact

Adding a temporary-impact term  $\frac{1}{2}\eta u_t^2$  modifies the FOC symbol to  $M(\xi) = c_\beta |\xi|^{\beta-1} + \eta/\gamma$ . The added constant provides high-frequency coercivity that the pure power-law symbol lacks, so  $u^\star \in L^2$  without spectral-decay assumptions on  $\alpha$ . Wiener–Hopf factorization of  $M$  gives modified one-sided factors; the crossover frequency  $\xi_* = (\gamma c_\beta / \eta)^{1/(1-\beta)}$  separates a long-memory fractional regime ( $|\xi| \ll \xi_*$ ) from a myopic signal-following regime ( $|\xi| \gg \xi_*$ ) in which  $u^\star \approx \alpha/\eta$ . The  $\eta \rightarrow 0$  limit is singular but recovers (18) under the spectral-decay hypothesis of §2.1.

## 5.3 Multi-asset extension

Empirical cross-impact between correlated assets — trades in one name moving others in the same principal-component sector — motivates a portfolio version of Theorem 1. For a cross-impact kernel  $\mathbf{K}(t) = |t|^{-\beta} \mathbf{A}$  with  $\mathbf{A} = Q\Lambda Q^\top$  symmetric positive-definite, Theorem 1 diagonalizes in the eigenbasis of  $\mathbf{A}$ : the scalar fractional-derivative rule (18) applies independently to each principal-component alpha with eigenvalue prefactor  $\Lambda_{ii}^{-1}$ . The symmetric case covers the leading positive-definite parameterization; non-symmetric cross-impact requires a matrix Wiener–Hopf factorization of  $\mathbf{A}(\xi)$ , which exists under the same log-integrability hypothesis but may carry non-zero partial indices, in which case the factor is not a simple fractional integral and the scalar reduction fails (Gohberg and Krein 1970).

## 5.4 Numerical implementation

Fractional derivatives discretize to Toeplitz matrices. On a uniform grid of  $N$  points,  $D_\pm^\nu$  is a lower- or upper-triangular Toeplitz whose entries are generalized binomial coefficients from the expansion of  $(1 - z)^\nu$ . Evaluating (18) over the whole grid costs  $O(N \log N)$  via FFT, compared with  $O(N^2)$  Nyström inversion of the Fredholm equation. The finite-interval formula (20) adds diagonal multiplications by  $B_T$  and  $B_T^{-1}$ , preserving the  $O(N \log N)$  cost.

## 5.5 Role of the forecast curve

Theorem 1 makes explicit that the optimal rate at time  $t$  depends on the *entire* forecast curve  $\bar{\alpha}(t, \cdot)$ . The stationary time series of optimal rates against a given signal is a deterministic functional of the family of forecast curves, with two immediate practical consequences.

*Signal engineering.* The perfectly-anticipating (non-adapted) optimum  $\gamma^{-1}C^{-1}\alpha$  attains value  $V_{\text{ant}}(\alpha) = \frac{1}{2\gamma} \mathbb{E}\langle \alpha, C^{-1}\alpha \rangle = \frac{1}{2\gamma c_\beta} \int |\xi|^{1-\beta} S_\alpha(\xi) \frac{d\xi}{2\pi}$  (spectral density normalized by  $\mathbb{E}[\alpha_t^2] = \int S_\alpha d\xi/2\pi$ ). This is an upper bound on the adapted value  $\frac{1}{2\gamma} \mathbb{E}\|P_+C^{-1}\alpha\|^2$  actually attained at  $u^*$  — for an OU signal the two differ by the factor  $\sin(\pi\beta/2)$ . The bound weights the signal spectrum by  $|\xi|^{1-\beta}$ , a specification that differs from an  $R^2$  of the signal against future returns: two signals with identical  $R^2$  can differ arbitrarily in tradeable value once their timescales differ, and the fractional-order weighting quantifies what makes one forecast better than another for a given impact kernel.

*Forecast horizon.* The Marchaud representation of  $D_-^\nu$  averages forecast increments  $\bar{\alpha}(s, r) - \bar{\alpha}(s, s)$  over all future lags  $r > s$  with weight  $(r - s)^{-1-\nu}$ . Forecasts are required over all horizons; horizon truncation gives an error decaying like  $r_{\text{max}}^{-\nu}$  in the maximum forecast horizon.

## 5.6 Joint gain–risk–cost formulation

The joint gain–risk–cost problem inherits the same operator structure once the two frictions share a coordinate. Cost acts on the trading rate  $u$ ; a mean–variance holding penalty  $\frac{\lambda}{2} \mathbb{E} \int x_t^\top \Sigma x_t dt$  acts on the position  $x_t = \int_{-\infty}^t u_s ds$ , which in rate coordinates is the operator with Fourier symbol  $\lambda \Sigma / \xi^2$ . Adding it to (1) replaces  $C$  by the symbol  $\gamma \hat{C}(\xi) + \lambda \Sigma / \xi^2$  — still a positive Toeplitz operator, admitting Wiener–Hopf and Gohberg–Krein factorization, to which Lemma 1 applies without modification. The pure trading limit ( $\lambda \rightarrow 0$ ) and the pure holding limit ( $\gamma \rightarrow 0$ ) are the two boundaries of this joint factorization.

## 6 Limitations

The results assume: (i) a symmetric positive-definite propagator kernel, so the cross-impact matrix in §5.3 must be symmetric; (ii) linear cost in the trading rate, with no bid–ask spread, tick-size effects, or non-linear temporary impact beyond the quadratic penalty of §5.2; (iii) constant volatility — volatility clustering and jumps in  $\alpha$  are excluded; (iv) linear KKT constraints only, ruling out inequality position bounds; (v) a stationary signal spectrum in the whole-line formulas of §4.2, with the OU case requiring  $\beta > 1/2$ . The interior error bound (Proposition 3) is pointwise in  $t$  and uses generic constants; sharp constants in  $\beta$  are not tracked.

## 7 Concluding Remarks

This paper establishes a closed-form solution to the gain–cost trading problem under a general symmetric positive-definite propagator kernel. The construction rests on the causal–anticausal factorization of the cost operator — Wiener–Hopf on the whole line, Gohberg–Krein on a finite interval, both realized as instances of the Arveson outer factorization for positive operators on a continuously nested Hilbert space. The adapted optimal rate is expressed as the composition of the causal square-root inverse of the cost operator with the adapted projection of the anticausal square-root inverse applied to the forecast curve, an expression whose form depends on the kernel only through its factorization.

For the power-law kernel, the whole-line factors are the Liouville (Weyl) fractional integrals of order  $\nu = (1-\beta)/2$ , and the closed form expresses the optimal rate as a composition of an anticausal and a causal Marchaud fractional derivative of order  $\nu$ , separated by the adapted projection  $P_+$ . Without  $P_+$  the composition would reduce to the symmetric Riesz derivative of order  $1 - \beta$ ; the projection preserves the two half-order pieces and encodes the causality constraint. On a finite horizon the fractional integrals acquire terminal-anchored endpoint-weight conjugations induced

by the reflected Gohberg–Krein factorization, and the optimal rate takes the form of a boundary-deformed fractional derivative that converges to the whole-line formula in the interior at rate  $O(d(t)^{-\nu})$ .

## A Appendix

**Proof of Lemma 1 (whole-line case).** With  $C_{\pm} = c_{\beta}^{1/2} I_{\pm}^{\nu}$  on  $L^2(\mathbb{R})$ ,  $C_{\pm}$  are Hilbert-space isomorphisms between  $\dot{H}^{-\nu}$  and  $L^2$  (respectively  $L^2$  and  $\dot{H}^{\nu}$ ) by Plancherel and the symbol  $c_{\beta}^{1/2} |\xi|^{(\beta-1)/2}$ . Their adjointness  $C_+^* = C_-$  follows from the kernel-flip  $(t-s)^{\nu-1} \mathbf{1}_{s \leq t} \mapsto (s-t)^{\nu-1} \mathbf{1}_{t \leq s}$ . On  $L_{\text{adap}}^2 \cap \dot{H}^{-\nu}$  the composition  $P_+ C P_+$  is bounded, symmetric, and strictly positive (from  $\hat{C} > 0$  on  $\xi \neq 0$ ), hence boundedly invertible on its range. Causality of  $C_+$  implies  $P_+^{\perp} C_+ P_+ = 0$ , so  $C_+$  preserves  $L_{\text{adap}}^2$  and  $C_+^{-1}$  commutes with  $P_+$  on adapted inputs. The adjoint identity  $P_+ C_- P_+^{\perp} = 0$  follows. Taking adjoints of  $C_+^{-1}: L^2 \rightarrow \dot{H}^{-\nu}$  gives  $P_+ C_-^{-1} P_+^{\perp} = 0$  as well, so  $C_-^{-1}$  preserves  $L_{\text{adap}}^2$ . For adapted  $u$ :  $P_+ C P_+ u = P_+ C_- C_+ u$ ; applying  $C_+^{-1} P_+ C_-^{-1}$  returns  $u$  via  $P_+ C_-^{-1} P_+ C_-$  acting as  $P_+$  on the adapted vector  $C_+ u$ .

**Proof of Lemma 1 (finite-interval case).** Take  $C_+ = R T^* R$  the reflected causal Gohberg–Krein factor of Proposition 2 and  $C_- = C_+^* = R T R$ , so that  $C_- C_+ = R(T T^*) R = R G_T R = G_T$ . Volterra causality of  $C_+$  (kernel on  $\{s \leq t\} \cap [0, T]^2$ ) gives  $P_+^{\perp} C_+ P_+ = 0$  and  $P_+ C_- P_+^{\perp} = 0$ ; the composition argument of the whole-line case then applies verbatim.

**Existence of the outer factor.** The Wiener–Hopf factorization on  $\mathbb{R}$  under the log-integrability hypothesis is classical (Wiener and Hopf 1931, Krein 1962). The finite-interval outer factorization  $G_T = T T^*$  is Arveson’s theorem for positive operators on a continuous nest (Arveson 1975, Thm 4.4.2); an equivalent constructive proof for symmetric weakly-singular integral kernels via spectral square root is in (Porter and Stirling 1990, Ex. 6.2/9.2); the explicit left-anchored factor (16) for the power-law kernel is derived in (Forde et al. 2022, pp. 590–591), and the reflected causal factor (17) required here follows from it by the reflection  $R$ .

**Proof of Corollary 1.** Define the candidate  $u_t^{\text{cand}} := \gamma^{-1} c_{\beta}^{-1} (D_+^{\nu} \zeta)(t)$ . Adaptedness:  $D_+^{\nu}$  at time  $t$  depends only on  $\{\zeta_s\}_{s \leq t}$ , and each  $\zeta_s \in \mathcal{F}_s \subset \mathcal{F}_t$  by (13) and  $\mathcal{F}_s$ -measurability of the forecast curve. FOC verification proceeds in three steps.

*Step (a).* Conditional Fubini (Klenke 2014, Thm 14.16) applied to the Marchaud representation gives  $\mathbb{E}_t[(D_+^{\nu} \zeta)(v)] = (D_+^{\nu} \hat{\zeta}_t)(v)$  with  $\hat{\zeta}_t(s) := \mathbb{E}_t[\zeta_s]$ .

*Step (b).* For  $s \leq t$ :  $\zeta_s \in \mathcal{F}_s \subset \mathcal{F}_t$ , so  $\hat{\zeta}_t(s) = \zeta_s$ . For  $s > t$ : by tower and conditional Fubini applied at conditioning time  $t$ ,  $\hat{\zeta}_t(s) = (D_-^{\nu} \bar{\alpha}(t, \cdot))(s)$ .

*Step (c).* The symbol identity  $\hat{C}(\xi)(-i\xi)^{\nu} = c_{\beta}(i\xi)^{-\nu}$  yields the operator identity  $C D_+^{\nu} = c_{\beta} I_-^{\nu}$  on  $L^2(\mathbb{R})$ . Substituting:

$$\gamma \mathbb{E}_t[(C u^{\text{cand}})(t)] = (I_-^{\nu} \hat{\zeta}_t)(t) = (I_-^{\nu} D_-^{\nu} \bar{\alpha}(t, \cdot))(t) = \bar{\alpha}(t, t) = \alpha_t,$$

using  $I_-^{\nu} D_-^{\nu} = \text{id}$  on  $H^{\nu}(\mathbb{R})$  (Samko et al. 1993, §5.3 Thm 5.3), applied to the future-half-line piece of the forecast curve, which is in  $H^{\nu}$  pathwise under the standing spectral hypothesis, and the fact that  $I_-^{\nu}$  at  $t$  samples only  $\hat{\zeta}_t(s)$  for  $s \geq t$ , where Step (b) gives the closed form.

*Uniqueness.*  $\hat{C}(\xi) = c_{\beta} |\xi|^{\beta-1} > 0$  on  $\xi \neq 0$ , so the quadratic penalty in (1) is strictly convex on  $L_{\text{adap}}^2$  and the adapted FOC has a unique solution.

*Admissibility.* The spectral hypothesis ensures  $|\xi|^{1-\beta}\widehat{\alpha}(\xi) \in L^2$  pathwise, so  $u^* \in L^2_{\text{adap}}$  by Plancherel.  $\square$

**Proof of Corollary 2.** The anticausal Marchaud derivative  $D_-^\nu$  in (20) acts on  $\alpha^{\text{eff}}$  extended by zero outside  $[0, T]$ , the natural convention for the finite-interval Volterra calculus (Samko et al. 1993, §13.5). Substitute the reflected factors (17) into (12):  $u^{*,T} = \gamma^{-1} C_+^{-1} P_+ C_-^{-1} \alpha^{\text{eff}}$  with  $C_+^{-1} = c_\beta^{-1/2} B_T D_+^\nu B_T^{-1}$  and  $C_-^{-1} = c_\beta^{-1/2} B_T^{-1} D_-^\nu B_T$  (inverting (17) using  $(I_\pm^\nu)^{-1} = D_\pm^\nu$ ). Collecting the two  $c_\beta^{-1/2}$  factors into  $c_\beta^{-1}$  gives (20).  $C_+^{-1}$  is a Volterra operator between the appropriate weighted Sobolev spaces (Samko et al. 1993, §13.5), not on  $L^2$  directly. Adaptedness follows from causality of  $C_+^{-1}$  (Volterra inverse of Volterra is Volterra).  $\square$

**Proof of Proposition 3.** The Marchaud tail bound:  $|(D_-^\nu f)(s) - (D_-^{\nu,[0,T]} f)(s)| \leq \frac{2\|f\|_\infty}{\Gamma(1-\nu)} (T-s)^{-\nu}$  from

$$(D_-^\nu f)(s) = \frac{\nu}{\Gamma(1-\nu)} \int_0^\infty \frac{f(s) - f(s+r)}{r^{1+\nu}} dr,$$

truncated at  $r = T - s$ . Symmetrically the outer  $D_+^\nu$  contributes a tail bounded by  $t^{-\nu}$ . The dominant interior truncation is on the closer boundary, giving the first term of (21). The terminal-weight conjugation contributes a further error: on the Marchaud support the ratio  $B_T(t)/B_T(s) = ((T-s)/(T-t))^\nu = 1 + O((t-s)/(T-t))$ , so its deviation from the whole-line unweighted operator is  $O(d(t)^{-1})$ , subdominant to  $d(t)^{-\nu}$  for  $\nu < 1$ . The endpoint-constraint correction enters through the Söhlgen–Tricomi eigenfunction  $\phi_1(t) = [t(T-t)]^{(\beta-1)/2}$  of the no-signal solution (Gatheral et al. 2012, Forde et al. 2022), which satisfies  $|\phi_1(t)| \leq d(t)^{-\nu} (T/2)^{-\nu}$ ; under  $\alpha^{\text{eff}} \in \dot{H}^\nu$  its coefficient is  $O(\|\alpha^{\text{eff}}\|_{\dot{H}^\nu} T^{-\nu})$ , giving the second term of (21).  $\square$

**Acknowledgements.** The author thanks Antoine Bambade, Raphael Benichou, Iacopo Mastromatteo, Jean-Yves Audibert, and Raphael Maillet for useful discussions. Drafting, literature review, and LaTeX preparation were assisted by Claude Opus via the Feynman research harness; all mathematical claims and results were verified by the author.

## References

- E. Abi Jaber and E. Neuman. Optimal liquidation with signals: The general propagator case. *Mathematical Finance*, 2025. arXiv:2211.00447.
- E. Abi Jaber, N. De Carvalho, and H. Pham. Trading with propagators and constraints: Applications to optimal execution and battery storage. *arXiv preprint arXiv:2409.12098*, 2024a.
- E. Abi Jaber, E. Neuman, and S. Tuschmann. Optimal portfolio choice with cross-impact propagators. *arXiv preprint arXiv:2403.10273*, 2024b.
- R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–39, 2001.
- W. Arveson. Interpolation problems in nest algebras. *Journal of Functional Analysis*, 20:208–233, 1975.
- J.-P. Bouchaud, Y. Gefen, M. Potters, and M. Wyart. Fluctuations and response in financial markets: The subtle nature of ‘random’ price changes. *Quantitative Finance*, 4:176–190, 2004.

- Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.
- A. Chakrabarti and A. J. George. A formula for the solution of general Abel integral equation. *Applied Mathematics Letters*, 7:87–90, 1994.
- M. Forde, L. Sánchez-Betancourt, and B. Smith. Optimal trade execution for Gaussian signals with power-law resilience. *Quantitative Finance*, 22:585–596, 2022.
- J. Gatheral. No-dynamic-arbitrage and market impact. *Quantitative Finance*, 10:749–759, 2010.
- J. Gatheral, A. Schied, and A. Slynko. Transient linear price impact and Fredholm integral equations. *Mathematical Finance*, 22:445–474, 2012.
- I. C. Gohberg and M. G. Krein. *Theory and Applications of Volterra Operators in Hilbert Space*. American Mathematical Society, 1970.
- T. Hytönen, J. van Neerven, M. Veraar, and L. Weis. *Analysis in Banach Spaces, Vol. I*. Springer, 2016.
- P. Jusselin and M. Rosenbaum. No-arbitrage implies power-law market impact and rough volatility. *Mathematical Finance*, 30:1309–1336, 2020.
- A. Klenke. *Probability Theory: A Comprehensive Course*. Springer, 2 edition, 2014.
- M. G. Krein. Integral equations on a half-line with kernel depending upon the difference of the arguments. *American Mathematical Society Translations (2)*, 22:163–288, 1962.
- C.-A. Lehalle and E. Neuman. Incorporating signals into optimal trading. *Finance and Stochastics*, 23:275–311, 2019.
- F. Lillo, J. D. Farmer, and R. N. Mantegna. Master curve for price-impact function. *Nature*, 421:129–130, 2003.
- H. Markowitz. Portfolio selection. *Journal of Finance*, 7:77–91, 1952.
- R. C. Merton. An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7:1851–1872, 1972.
- E. Neuman and M. Voß. Optimal signal-adaptive trading with temporary and transient price impact. *SIAM Journal on Financial Mathematics*, 13:551–575, 2022.
- A. A. Obizhaeva and J. Wang. Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets*, 16:1–32, 2013.
- D. Porter and D. S. G. Stirling. *Integral Equations: A Practical Treatment from Spectral Theory to Applications*. Cambridge University Press, 1990.
- S. G. Samko, A. A. Kilbas, and O. I. Marichev. *Fractional Integrals and Derivatives: Theory and Applications*. Gordon and Breach, 1993.
- N. Wiener. *Extrapolation, Interpolation and Smoothing of Stationary Time Series*. MIT Press, 1949.
- N. Wiener and E. Hopf. Über eine klasse singulärer integralgleichungen. *Sitzungsberichte der Preussischen Akademie der Wissenschaften Berlin*, pages 696–706, 1931.